

Example 1 — Integers and identifiers

Create a file called scanner.l and paste the following.

```
%option noyywrap

%{
#include <stdio.h>
%}

%%

[0-9]+ { printf("INTEGER: %s\n", yytext); }

[a-zA-Z][a-zA-Z0-9]* { printf("IDENTIFIER: %s\n", yytext); } [

\t\n]+ { /* ignore whitespace */ }

. { printf("UNKNOWN: %s\n", yytext); }

\n {return 0;}

%%

int main()
{
    yylex();
    return 0;
}
```

Test input

```
age 25
student1 2026
total 500
```

Example 2 — Keywords and operators

```
%option noyywrap
```

```

%{
#include <stdio.h>
%}

%%

"int" { printf("KEYWORD: INT\n"); } "if" {
printf("KEYWORD: IF\n"); }
"else" { printf("KEYWORD: ELSE\n"); } "return" {
printf("KEYWORD: RETURN\n"); }

[0-9]+ { printf("INTEGER: %s\n", yytext); }

[a-zA-Z][a-zA-Z0-9]* { printf("IDENTIFIER: %s\n", yytext); }

"=" { printf("ASSIGN: %s\n", yytext); } "+" {
printf("PLUS: %s\n", yytext); } "-" { printf("MINUS:
%s\n", yytext); } "*" { printf("MULTIPLY: %s\n",
yytext); } "/" { printf("DIVIDE: %s\n", yytext); }

[ \t\n]+ { }

. { printf("UNKNOWN: %s\n", yytext); } %%

int main()
{
    yylex();
    return 0;
}

```

Example 3 — Scan a small program

```

">=" { printf("GREATER_EQUAL: %s\n", yytext); }

"(" { printf("LEFT_PAREN: %s\n", yytext); } ")" {
printf("RIGHT_PAREN: %s\n", yytext); }

"{" { printf("LEFT_BRACE: %s\n", yytext); } "}" {
printf("RIGHT_BRACE: %s\n", yytext); }

";" { printf("SEMICOLON: %s\n", yytext); }

```

Example 4: Create a scanner.l file and add the following code

```
/* Definitions */
%{
#include <stdio.h>

int keywords = 0;
int identifiers = 0;
int numbers = 0;
int operators = 0;
int special_characters = 0;
}%

/* Rules and Actions */
%%
"int"|"float"|"char"|"if"|"else"|"while"|"return" { keywords++; }

[a-zA-Z_][a-zA-Z0-9_]* { identifiers++; }

[0-9]+ { numbers++; }

"+"|"-"|"*"|"/"|"="|"!="|"<"|">" { operators++; }

[{ } ( ) ; , \n ] { special_characters++; }

[ \t\n]+ { /* Ignore whitespace */ }

. { printf("Unknown character: %s\n", yytext); }

%%

int yywrap()
{
    return 1;
}

int main()
{
    yyin = fopen("test.txt", "r");

    if (yyin == NULL)
    {
```

```

        printf("Could not open test.txt\n");
        return 1;
    }

    yylex();

    printf("\nLexical Analysis Results:\n");
    printf("-----\n");
    printf("Keywords : %d\n", keywords);
    printf("Identifiers : %d\n", identifiers);
    printf("Numbers : %d\n", numbers);
    printf("Operators : %d\n", operators);
    printf("Special characters : %d\n", special_characters);

    fclose(yyin);

    return 0;
}

```

Create a test.txt file and add the following code

```

int age = 20;
float salary = 50000;

if (age > 18) {
    salary = salary + 1000;
}

```